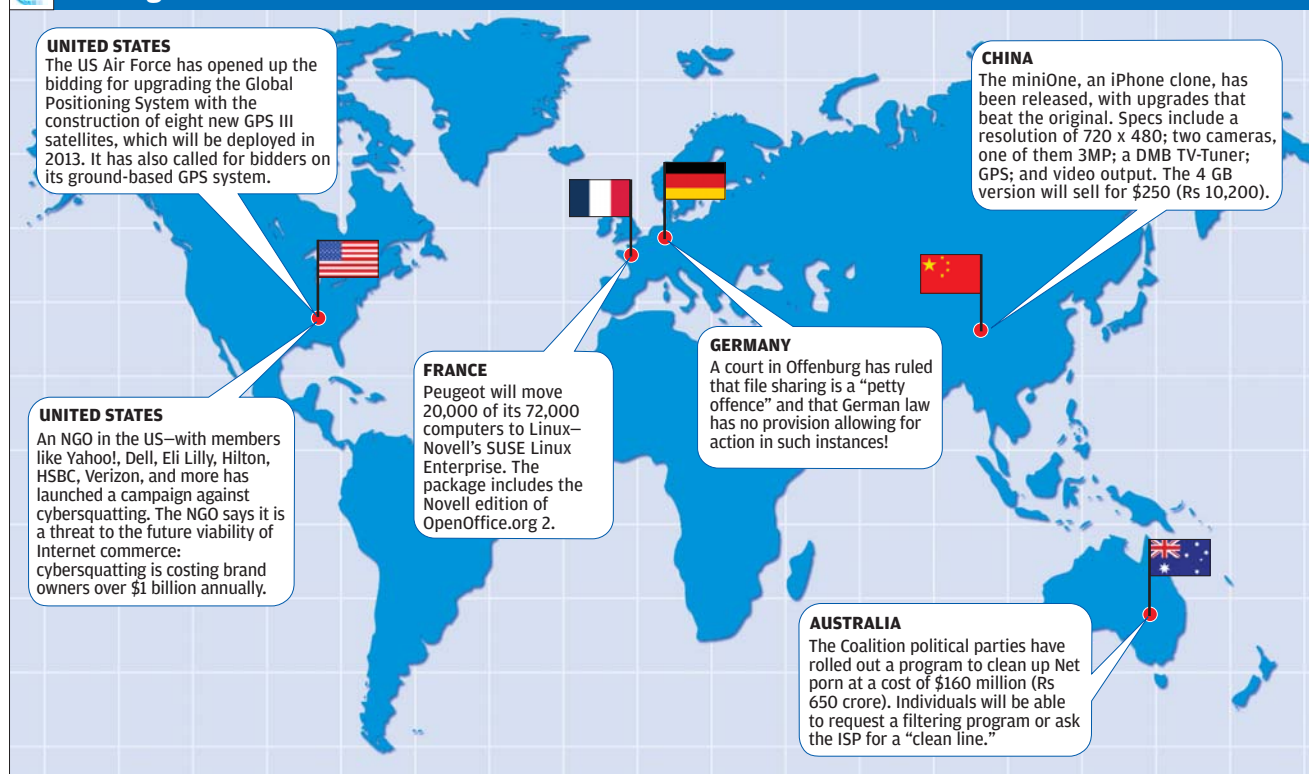




NOW, THE MORE THE MERRIER

Software Reprogrammed

The advantages of multi-core processors—of which a lot is being talked about these days—include the use of cache memory, and more importantly, distribution of threads for processing to the various processors. So that you get a clearer picture, here's what threads are: think of a thread as part of a program *in execution*. A program is software; when it runs, it creates (one or more) thread(s), each of which do part of the overall task. So, for example, if your browser is loading a page, the browser is the program, and there is one thread that loads an image, one that loads another, one that runs a script, and so on. (This is simplistic, mind you.)

The hardware advantages are meaningless without software to take advantage

of it. The majority of common software applications cannot as of now take use multi-core to its potential—they just aren't coded to use multiple concurrent threads in an effective manner.

"The power of multi-core processors to increase software performance is fantastic, but not automatic and not happening fast enough," said Elliot Garbus, general manager of Intel's Developer Relations Division. In July of this year, Intel introduced a global-scale curriculum for training developers in better multi-threaded application design. (It should be clear why Intel took the initiative.)

Now, AMD has made available a specification called "Light-Weight Profiling" (LWP), which developers can use to code multithreaded applications and improve their performance using the features to be offered in future AMD processors.

In simple words, LWP manages to look at the threads in a program,

collect this data, and report to the programmer.

The Java Runtime Environment (JRE) is probably installed on your computer to facilitate interaction of your browser (for example) with Web applications. (Online Java games need the JRE.) The JRE, as an example, will benefit from LWP; it is capable of running multiple threads, and it is widely used in application development. LWP can also be implemented in the development of operating systems and compilers.

On the other side of the battlefield, Intel recently published eight papers on *programmable multi-core processors*. These focus on memory architecture, cache sharing, on-die integration of cores, and *hardware* architectural changes that will accelerate application multithreading and parallel programming.

These efforts will in all likelihood close, to some extent, the gap between where the hardware and software are today. The developer community hasn't

yet reacted in regards to the viability of the LWP proposition, though.

The best use of these developments can be made by virtual machines (cf. virtualisation) and allied applications at the server level. However, the gestation period for software that can follow multiple instructions cannot but increase. Software production costs will rise—developers will need to be trained in new coding methods.

CHINKS IN THE CHIP

Bypassing The Port

The EU, along with the US and twelve other countries have started issuing e-passports, which involve biometric recognition; in some cases, only to particular officials and citizens, and in the case of the US, to all citizens. The International Civil Aviation Organisation (ICAO), a transnational organisation that sets the standards and